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G. L. BAJAJ INSTITUTE OF TECHNOLOGY & MANAGEMENT

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING – ARTIFICIAL INTELLIGENCE

PROJECT POSTER PRESENTATION

SESSION (2026-2027) BATCH – 2023

FINAL YEAR PROJECT

Capstone Evaluation

AI & SYSTEMS TRACK

GROUP ID: 40

PROJECT TITLE:

ResumeGrind: AI-Powered Adaptive Technical Interviewer & Resume Cross-Examination Engine

TEAM MEMBERS:

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ABSTRACT

- **Addresses Technical Hiring Inefficiencies:** Solves widespread resume exaggeration, proxy interview cheating, and generic, un-grounded mock interview simulators.
- **Autonomous Multi-Tier Cross-Examination:** Deep-parses candidate resumes and inspects live GitHub repositories (AST parsing, commit history) to generate targeted technical inquiries.
- **Live LeetCode-Style DSA Coding Sandbox:** Integrates a multi-language (Python, Java, C++, JS) browser-based execution sandbox with automated test case evaluation.
- **Acoustic Speech & Delivery Analytics:** Measures words per minute (WPM), speech pacing, and filler word frequency to coach holistic verbal delivery.
- **Zero Question Repetition & Command Control:** Employs memory-tracked historical deduplication across 15 senior engineering tracks and cockpit command directives.
- **Google XYZ Resume Enhancement:** Delivers post-session feedback with quantifiable metric refactoring suggestions for candidate resumes.

SYSTEM ARCHITECTURE & METHODOLOGY

1. Resume & GitHub Ingest

- PDF text extraction & OCR
- GitHub REST / GraphQL AST
- Commit logs & key dependencies
- Claim extraction & tokenization

2. Context & RAG Engine

- Dense semantic embeddings
- Project tech stack mapping
- Ground-truth verification
- Failure mode question synthesis

3. Live Cockpit & Personas

- WebSpeech API voice stream
- Audio level VU metering
- Cockpit COMMAND directives
- Gemini 3.5 Flash-Lite LLM

4. LeetCode Code Sandbox

- Isolated sandbox compiler
- Python, Java, C++, JavaScript
- Automated unit test assertion
- O(N) vs O(N²) algorithmic triage

Frontend Tier (React + Vite)
TailwindCSS, Monaco/CodeMirror Sandbox, WebSpeech SpeechRecognition & Synthesis, Framer Motion animations.

Backend Services (FastAPI)
Async Python 3.11, PyPDF2/pdfminer, GitHub Octokit AST analysis, Session Store with DynamoDB & in-memory fallback.

AI & Sandbox Orchestration
Gemini 3.5 Flash-Lite, Local Ollama fine-tuned Llama 3, Docker sandboxed subprocess runtime, Speech WPM analyzer.

Security & Reliability Controls:

- Sandboxed Subprocess Execution
- Zero Long-Term Voice Retention
- Deterministic Gibberish Guardrail
- AWS DynamoDB 24h TTL

EXPERIMENTATION & BENCHMARK RESULTS

- **Provider Latency:** Gemini 3.5 Flash-Lite achieved **1.82s** response time, enabling fluid conversational dialogue compared to Local Ollama (8.14s).
- **Question Uniqueness: 100% unique** follow-up questions across 15 technical domains with zero looping.
- **Algorithmic Triage:** Sandbox graded solutions in **< 140ms** with instant time-complexity feedback.

Evaluation Metric	Gemini Flash	Ollama Local	Target
Turn Response Latency	`1.82 s`	8.14 s	< 2.5 s
Question Deduplication	`100.0%`	100.0%	> 98.0%
DSA Compile & Run Time	`138 ms`	142 ms	< 250 ms
Gibberish Detection Recall	`100.0%`	96.2%	100.0%
Speech Pacing Accuracy	`95.8%`	95.8%	> 90.0%

CONCLUSION & IMPACT

- **Autonomous Technical Rigor:** Successfully developed an end platform that automates rigorous technical screening with contextual ir
- **Honest Skill Validation:** Dissects project source code to verify real h contributions, eliminating superficial buzzword inflation.
- **Bimodal Interview Paradigm:** Seamlessly blends high-level system cross-examination with live LeetCode algorithmic implementation cohesive session.
- **Actionable Feedback Loop:** Generates post-interview diagnostic featuring Google XYZ resume improvements and tailored code refactor
- **Scalable Deployment:** Built on a serverless-ready architecture cap handling concurrent student interview streams with zero perfo degradation.

End-to-End Interview Execution Workflow

1. **Resume & GitHub Ingest**
   Parses candidate PDF claims & GitHub AST repositories
2. **Claim & Tech Mapping**
   Identifies architecture claims, concurrency & database tools
3. **Multi-Turn Cross-Exam**
   Dynamic persona questioning targeting bottlenecks & trade-offs
4. **Live LeetCode Sandbox**
   In-browser IDE execution with O(N) automated test runner
5. **Analytics & Feedback**
   Speech delivery metrics & Google XYZ resume improvements